ENERGIA POWER · HOME LAB

Home Lab Hub How It Works

A plain-language guide to the technology behind the hub: what each piece does, how the pieces work together, and why each one matters.

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Prepared November 9, 2026. It describes the system as it runs today: the hub, its apps, the three-computer cluster and the home network. Passwords, keys and private addresses are deliberately left out.

1. The big picture in one page

Home Lab Hub is a private "front door" to a family business's software and to the house's cameras and smart devices. Sign in once at **hub.energiapower.llc** and you see a page of tiles, one per app, each with a green or red light that shows whether it's working.

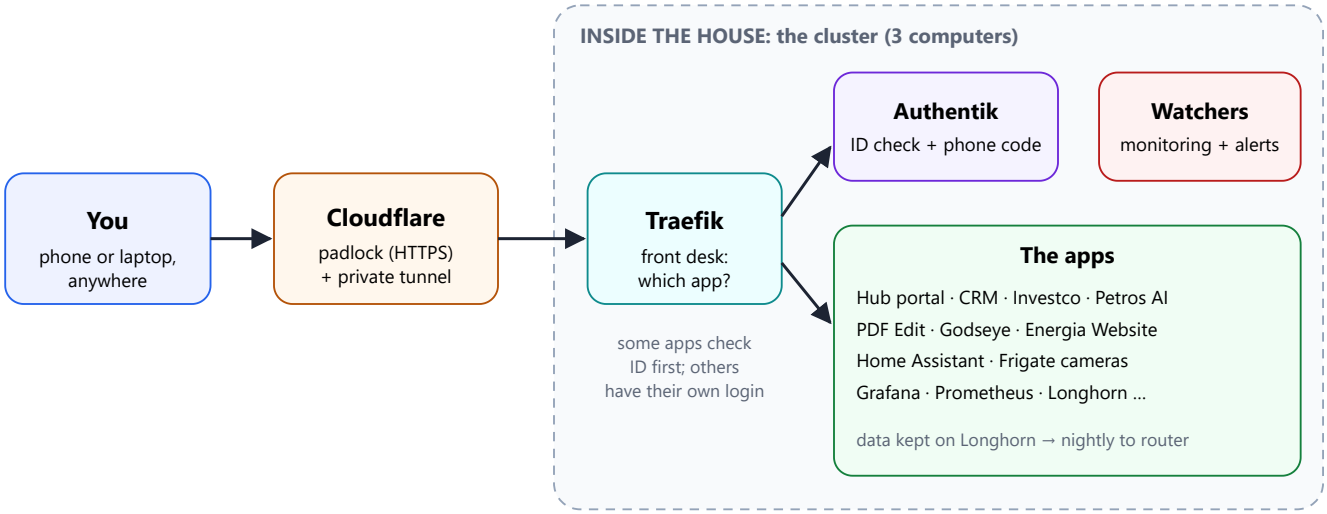


Figure 1. A visitor never connects to the house directly: everything comes in through Cloudflare's tunnel, then the front desk (Traefik) and, for protected apps, the ID check (Authentik).

Behind that simple page is a small, professional-grade setup that runs at home instead of in a paid cloud. The easiest way to picture it is an **apartment building**:

In the building…	In Home Lab Hub…
The land and the building	Three small computers in the house, joined into one "cluster", plus the home router with two 4 TB drives
The building manager	**Kubernetes (k3s)**: decides which computer runs each app, and restarts an app if it crashes
The apartments	The apps: CRM, Investco, Petros AI, Home Assistant and the rest
A private, guarded entrance from the street	**Cloudflare Tunnel**: the only way in from the internet. No door on the house is left open.
The front desk that sends visitors to the right apartment	**Traefik**: routes each web address to the right app
The security office that issues ID badges	**Authentik**: sign-in with a password plus a phone code
The lobby directory	**The hub portal** itself: the page of tiles

Blueprints, and a foreman who keeps the building matching them	**GitHub** holds every setting as text; **Argo CD** keeps the cluster matching it
Storage units and an off-site vault	**Longhorn** keeps app data on the cluster; nightly **backups** go to the router's drives
Smoke detectors and the alarm company	**Prometheus, Grafana, Loki and Alertmanager** watch everything and email when something's wrong
Doorbells, cameras and the light switches	**Home Assistant** and **Frigate** for cameras, lights and sensors

2. The physical equipment

Everything runs on a handful of devices in the house. No monthly cloud-server bill is involved, apart from small services such as email delivery and the domain name.

Device	What it is	Its job
openclaw	Small PC (Intel i5, 8 GB memory)	The cluster's "control plane": the computer the building manager works from. It also runs a few light apps.
worker-01 and **worker-02**	Small PCs (Intel i7, 16 GB memory, built-in Intel graphics)	Do most of the work. The graphics chip on worker-02 runs the camera AI (spotting people and cars). worker-02 also has a carefully fenced connection to the smart-device Wi-Fi.
Synology RT2600ac router	The home router, with two 4 TB drives attached	Connects the house to the internet. Keeps three separate networks (family, guests, smart devices) and stores the nightly backups.
Cameras	2 Reolink, 3 Blink (plus 6 not yet installed), 2 Wyze	Watch the garage, basement, parking lot, front door and porch. The Wyze Cam Pans aren't connected yet.

> **Why three computers instead of one?**
> If one computer fails or needs a restart, the building manager moves its apps to the others, and each app's data is already copied onto more than one machine. One big computer would be a single point of failure.

Three separate home networks

The router keeps the **family network**, the **guest network** and a **smart-device network** (cameras, plugs and similar) walled off from each other. Cheap smart devices are a common way in for attackers, so they live on their own network and can't reach the family's computers. The one deliberate exception, a connection from worker-02 for the Wyze cameras, has its own firewall that only accepts traffic from those two cameras.

3. The apps people actually use

These are the tiles on the hub today. **Audience** is who can see the tile: `family` means the family group and admins; `admin` means administrators only. The hub hides tiles a person isn't allowed to see.

App	Audience	What it's for	Sign-in
CRM (Energia)	family	The business's customer records, contracts, quotes, emails and tasks. Its data lives in its own database in the cluster, backed up nightly.	Own login
PDF Edit	family	Edit and sign PDF documents.	Own login
Investco	family	Investment tracking. Pulls market data from outside services such as Schwab and Massive.	Own login
Godseye	family	A business analysis app.	Own login
Petros AI	family	A supply-chain "digital twin": a simulated business where AI agents learn to make buying and shipping decisions. It includes a QA Lab where admins can run its automated tests.	Own login
Energia Website	family	The public business website.	Public
Home Assistant	family	The smart-home control panel: all cameras on one page, the parking lot light, alerts.	Own login + phone code
Frigate	family	The camera recorder with AI that spots people and cars.	Authentik
Argo CD	admin	Shows what's deployed and whether it matches the blueprints. Home network only.	Own login
Grafana	admin	Charts of the cluster's health, and the logs.	Own login
Prometheus	admin	The raw health measurements behind those charts.	Authentik
Alertmanager	admin	Sends the "something's wrong" emails.	Authentik
Authentik	admin	Manages users, groups, invites and sign-in rules.	Itself
Longhorn	admin	Storage, restores and deleting backups. Asks for a fresh phone code every time.	Authentik + fresh code

The hub also has admin-only pages: **Manage** to add or edit tiles (every change is recorded in an audit log), **Health** for each computer's load and each app's status, and **Backups** for when each app's data was last saved, with a "Back up now" button.

4. The foundation: running many apps on three computers

Each app is packaged as a **container**: a sealed box holding the app and everything it needs to run, so it behaves the same on any computer. Think of a shipping container: the crane doesn't care what's inside.

Kubernetes (k3s)

building manager

Spreads the containers across the three computers, restarts any that crash, and replaces one when a new version arrives. k3s is a lightweight edition suited to small computers.

Like a building manager who keeps every apartment occupied and fixes anything broken.

Calico

internal roads

The private network that lets apps talk to each other inside the cluster. It also enforces **network rules**, such as "only the CRM may talk to the CRM's database".

Like hallways with locked doors between wings.

MetalLB

street addresses

Gives some apps a fixed address on the home network, so devices in the house can find them.

Like assigning unit numbers.

Traefik

front desk

Receives every web visit and sends it to the right app by name, for example "frigate…" to the camera app. It can ask Authentik to check ID first.

Like a receptionist directing visitors.

Docker images on GitHub (GHCR)

warehouse

Finished app packages (container images) are stored in GitHub's container registry, each labelled with its version, for example Petros 2.4.4.

Like a warehouse of prefabricated apartments, each with a model number.

PostgreSQL

filing cabinets

The database most apps use: the hub, CRM, Investco and Authentik each have their own. Petros uses a hosted one (Supabase), with a nightly copy kept at home.

Like each business keeping its own locked filing cabinet.

Why it matters

Without Kubernetes, someone would have to start every app by hand, notice crashes and restart them, and rebuild everything after a computer failure. With it, the cluster heals itself: a crashed app is back in seconds, usually before anyone notices.

5. Getting in safely from the internet

Most home setups "open a port" on the router so the internet can reach a computer inside. That's like leaving a side door unlocked. Home Lab Hub doesn't do this.

Cloudflare

guarded entrance

Owns the public face of every address ending in **energiapower.llc**. It provides the padlock (HTTPS encryption), blocks obvious attacks, and hides the house's real internet address.

Cloudflare Tunnel

private corridor

A small program inside the cluster dials *out* to Cloudflare and keeps that line open. Visitors' traffic comes back down this line, so the router never has to accept connections from strangers.

Each public address (hub, crm, ai, home, frigate…) is a "route" in the Cloudflare dashboard that says which app inside should receive it. Admin-only tools such as Argo CD have no public route at all, so they can only be reached from inside the house.

> **Why it matters**
>
> The router has no open doors, and the house's address isn't published. An attacker scanning the internet finds nothing to knock on. Everything they could reach goes through Cloudflare first, then through a login.

6. Sign-in and who can see what

Authentik

ID office

The central sign-in service at **auth.energiapower.llc**. It checks a password *and* a 6-digit code from a phone app (two-factor sign-in). It decides which **groups** a person belongs to: administrators, **hub-family** (all family apps), or **hub-limited** (fewer apps). New people join by invitation only.

Single sign-on

one badge, many doors

Signing in to the hub also signs you in to the protected tools (Frigate, Prometheus, Alertmanager, Longhorn) without typing your password again. The most dangerous tool, Longhorn, which can delete backups, asks for a fresh phone code every time anyway.

Two kinds of apps

Protected by Authentik

Frigate, Prometheus, Alertmanager and Longhorn have no login of their own. Traefik asks Authentik "is this person allowed?" before letting anyone near them.

Their own login

CRM, Investco, Petros, PDF Edit, Godseye, Grafana and Home Assistant already have proper logins (and the phone app for Home Assistant). The hub simply links to them, to avoid signing in twice.

Why it matters

A stolen password alone isn't enough to get in, because the phone code is also needed. Each family member sees only the tiles meant for them. And there is one place to remove someone's access, instead of a dozen separate accounts.

7. From a code change to a live app

Every app and every setting is written down as text in **GitHub** (the "blueprints"). Nobody changes the live system by hand. They change the blueprint, and machines do the rest. This is called **GitOps**.

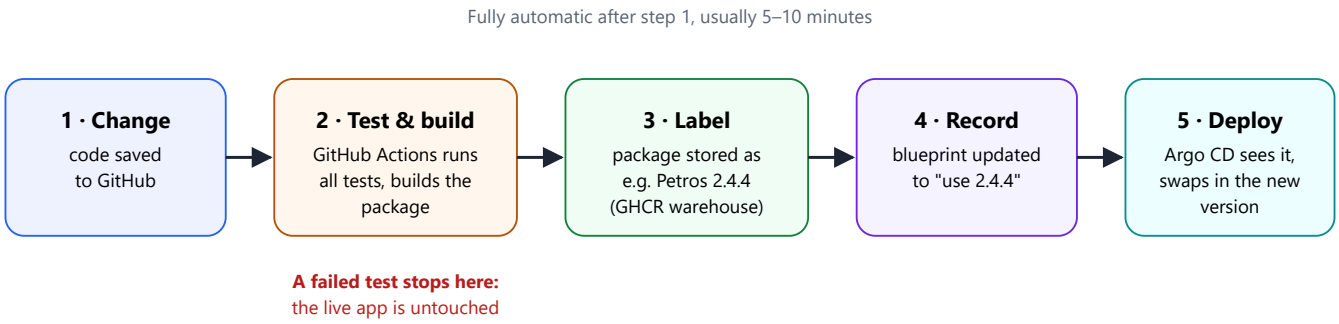


Figure 2. The Petros release line. Each passing change gets the next version number (2.4.1, 2.4.2 …), which shows on its login page.

GitHub

blueprints

Keeps every version of every file, so any change can be seen, explained, and undone.

GitHub Actions

quality gate

Runs each app's automated tests on every change. For Petros these are 15 simulation-engine checks plus browser tests driven by **Playwright**, a robot that clicks through the site like a person.

Argo CD

foreman

Constantly compares the cluster with the blueprints. If someone changes something by hand, it puts it back; if the blueprint changes, it applies it.

Sealed Secrets

locked envelopes

Passwords can't be stored in plain text on GitHub. They are sealed with a key that only the cluster has, so the blueprints can be shared safely.

Renovate

update reminders

Watches for new versions of the building blocks and suggests updates to the hub, so security fixes aren't missed.

Security scanners

inspectors

Trivy (known weaknesses in packages), Gitleaks (passwords accidentally saved) and Semgrep (risky code patterns) check the hub's code before it ships.

Why it matters

Changes are tested before they go live, every change is recorded and can be reversed, and the whole system could be rebuilt from the blueprints if the hardware were lost.

8. Keeping data safe: storage and backups

Apps come and go, but their data (customer records, contracts, settings, camera recordings) has to survive.

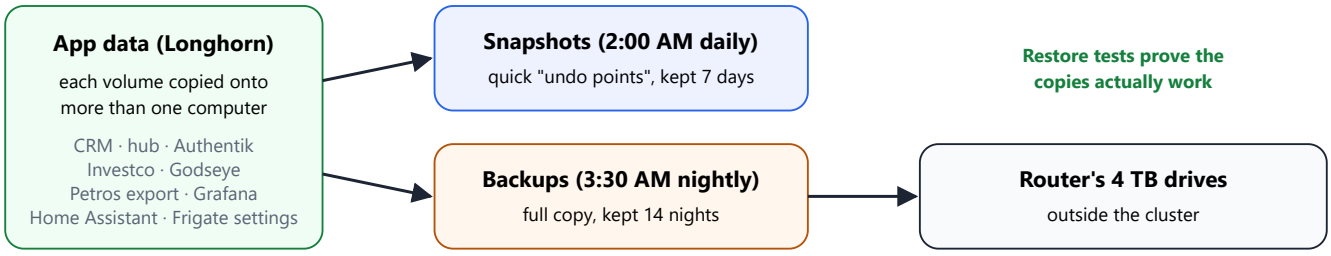


Figure 3. Two layers of protection: quick undo points on the cluster, and a nightly copy on separate hardware.

Longhorn

storage units

Gives each app a "volume" (its disk) and keeps copies of it on more than one computer, so a failed computer loses nothing. It also takes the snapshots and backups.

Nightly backups

off-site vault

Every night at 3:30 the important volumes are copied to the router's drives. The hub's Backups page shows when each app was last saved, and an email alert goes out if a backup fails or is overdue.

Petros nightly export

copy of a cloud app

Petros keeps its database with a hosted service (Supabase) and its trained AI models with Google Cloud. Every night a job copies both to a volume at home, which then goes into the backups as well.

Restore tests

fire drills

A backup is only useful if it restores. Scripts restore the newest backup into a separate test area and compare it with the live data. The CRM test on Nov 9 brought back all 45 tables; the only differences were rows added since 3:30 that morning.

Still to do

Today every backup sits in the house. A fire or theft could take the computers and the router together, so an off-site copy (another location or cloud storage) is planned for the hardening phase.

9. Watching over everything: monitoring and alerts

Prometheus

sensors

Every few seconds, records measurements from every computer and app: memory, disk space, whether each app is running, whether backups succeeded.

Alertmanager

alarm company

When a measurement crosses a line (disk nearly full, an app down, a backup overdue), it sends an email.

Grafana

dashboard

Turns the measurements into charts, for example memory use per computer over the last week.

Loki

diary

Collects every app's log messages in one place, searchable from Grafana. This is how problems like "why did the camera stop?" get diagnosed.

The hub's own **status lights** and **Health** page give the everyday view: green if an app answers, red if it doesn't, plus each computer's load. Prometheus and Grafana are for digging deeper.

> **Why it matters**
>
> Problems get noticed by a machine, often before a person would. And when something does break, the logs and measurements show what happened instead of leaving people to guess.

10. Cameras and the smart home

Cameras from three brands each work differently. The goal is one place to see them all, recordings when something happens, and alerts that matter.

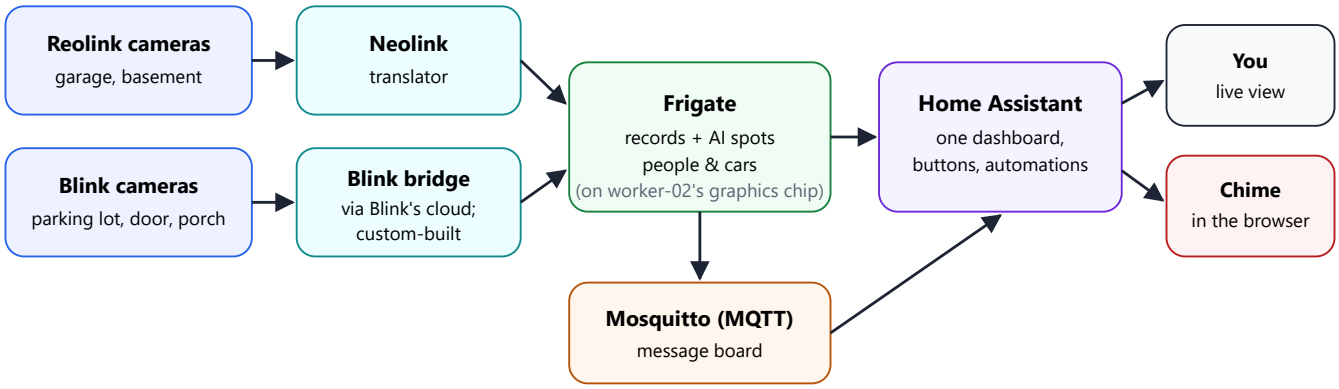


Figure 4. Every camera ends up in Frigate (recording and detection) and in Home Assistant (one dashboard and controls).

Home Assistant

control panel

The smart-home app at **home.energiapower.llc**: all cameras on one page, the parking lot light button, and automations such as "chime when someone's in the parking lot". It has its own login with a phone code, and works with the phone app.

Frigate

recorder with eyes

Records the cameras around events and uses AI (OpenVINO on an Intel graphics chip) to recognise people and cars. It keeps clips for a limited time so the disk never fills up.

Neolink

translator

The Reolink E1 cameras speak their own language; Neolink translates it into the standard camera format (RTSP) and passes pan/tilt commands back.

Blink bridge

custom-built

Blink cameras only stream through Blink's cloud. This bridge, written for this project, starts live views on request. The battery cameras only wake when someone asks, to save battery. The wired parking lot camera stays live, and the bridge passes the video to Frigate through **go2rtc**, a video relay.

Mosquitto (MQTT)

message board

A tiny message service the camera tools use to talk to Home Assistant: "person detected", "start live view", "light on".

Browser Mod

speaker in the browser

An add-on that lets Home Assistant play the parking-lot chime in a computer's browser while the dashboard is open, with an on/off bell button.

Not working yet: the Wyze Cam Pans (they don't answer the bridge built for them). The Blink doorbell's live view depends on its weak Wi-Fi connection.

11. Four everyday stories, start to finish

A. Opening the CRM from a coffee shop

1. You type **hub.energiapower.llc**. Cloudflare answers with the padlock and passes the visit down the private tunnel.
2. Traefik sees "hub" and hands the visit to the hub portal, which sends you to Authentik to sign in: password, then the 6-digit code from your phone.
3. The hub shows the tiles your group may see, with green lights because each app answered its health check seconds ago.
4. You click **CRM**. It opens crm.energiapower.llc, which arrives through the same tunnel. The CRM has its own login and reads your customers from its database in the cluster.

B. A car pulls into the parking lot

1. The Blink camera's live video flows through Blink's cloud to the bridge, and on to Frigate.
2. Frigate's AI spots a car and posts "car in parking lot" on the MQTT message board.
3. Home Assistant sees it. Because the bell is switched on, it tells Browser Mod to play the chime and show "Car in the Parking Lot" in every registered browser.
4. Frigate keeps a clip of the event. You can tap the Parking Lot card to watch live, and switch the light on.

C. A Petros improvement ships

1. A change to Petros is saved to GitHub.
2. GitHub Actions builds the app and runs every test inside it. One failure, and it stops there, with nothing changed.
3. If all pass, the package is stored as the next version (for example 2.4.4) and the blueprint is updated to use it.
4. Argo CD notices within minutes and swaps the old version for the new one. The login page now shows **v2.4.4**.

D. 3:30 AM, every night

1. Longhorn copies each important volume to the router's drives: the CRM, hub, Authentik and Investco databases, Godseye's accounts, the Petros export, Home Assistant, Grafana, and Frigate's settings. Camera recordings aren't included, because they expire after a few days anyway; important clips will be kept on the router (Phase 6).

2. Prometheus records whether each backup worked. If one fails, or a backup is overdue, Alertmanager emails the admin.
3. In the morning, the hub's Backups page shows the time of every last successful save.

12. Why each piece matters (summary)

Technology	In one sentence	Works closely with	Without it…
Kubernetes (k3s)	Runs and heals every app on three computers	Everything below	Apps started and fixed by hand; one dead computer means an outage
Calico	Private network and traffic rules between apps	Kubernetes	Any app could talk to any database
MetalLB	Fixed home-network addresses for some apps	Traefik, Home Assistant	Devices in the house couldn't find the apps
Traefik	Sends each web address to the right app	Cloudflare Tunnel, Authentik	Every app would need its own entrance
Cloudflare + Tunnel	Safe way in from the internet, with the padlock	Traefik	Open doors on the router, and a public home address
Authentik	One sign-in with a phone code; groups decide access	Hub, Traefik, Frigate, Longhorn	Many separate passwords; no central way to remove access
Hub portal	The page of tiles, status lights, health and backups	Authentik, Postgres, Prometheus, Longhorn	No single place to find and check everything
PostgreSQL	The apps' databases	Longhorn	Nowhere reliable to keep business records
GitHub + Actions + GHCR	Blueprints, automatic testing, versioned packages	Argo CD	Untested changes; no history; no way to undo
Argo CD	Keeps the cluster matching the blueprints	GitHub, Kubernetes	Manual deployments and settings drift over time
Sealed Secrets	Passwords stored safely alongside the blueprints	Argo CD	Passwords in plain text, or kept off the record
Longhorn	App disks copied across computers; snapshots and backups	Router drives, Prometheus	A disk failure loses data; no nightly copies
Prometheus, Grafana, Loki, Alertmanager	Measure, chart, log and alert	Every app; email	Problems found only when someone complains
Home Assistant	One smart-home dashboard and automations	Frigate, MQTT, Browser Mod	A separate app for every brand of device

Frigate	Records cameras; spots people and cars	Neolink, Blink bridge, MQTT	No recordings or detections for the Reolink and wired cameras
Neolink / Blink bridge / go2rtc	Translate each brand's video into a standard format	Frigate	Those cameras couldn't join the shared view
Mosquitto (MQTT)	Message board between camera tools and Home Assistant	Frigate, bridges, Home Assistant	No detection events, buttons or light control
Synology router	Internet, separate networks, backup storage	Longhorn, everything	Smart devices mixed with family computers; no backup target

13. What's next

The project is built in phases. Phases 0–3 are finished (foundation, sign-in, the portal, monitoring and backups). Phase 4 (cameras and devices) is mostly done.

Phase	What it adds	Status
4 · Devices & cameras	Remaining: the Wyze cameras, more smart devices (plug, chiller, bulbs, thermostat, locks, garage door, solar), and copying important clips to the router.	Mostly done
5 · Computers	Reach the family's PCs from the hub in a web browser (remote desktop), behind the same sign-in and phone code.	Planned
6 · Router storage	Use the router's two 4 TB drives from the hub: browse files, see space and drive health, and give apps a shared folder (also where important camera clips will go).	Planned (suggested next)
7 · Hardening	A security pass: an off-site copy of the backups, tighter rules between apps, password rotation, proper certificates inside the cluster, and relief for the busy control-plane computer.	Planned

> **Related projects**
>
> MLflow for tracking Petros's AI training experiments, and an energy-broker AI agent that turns requests for proposals into contracts. Both would run on this same platform and appear as tiles on the hub.

14. Glossary

App / service
A program people use through a web browser, such as the CRM.

Backup
A copy of data kept somewhere else, to bring it back after a loss.

Cluster
Several computers working together as one system.

Container
An app packed with everything it needs, so it runs the same anywhere.

Control plane
The part of the cluster that makes the decisions; here, the computer "openclaw".

Database
Organised storage for records, such as customers and contracts.

Deploy
Put a new version of an app into use.

DNS / domain
The internet's address book; energiapower.llc is the domain.

Encryption (HTTPS)
Scrambling traffic so only the two ends can read it; the browser's padlock.

GitOps
Running a system by changing written blueprints that machines then apply.

Image
A finished, versioned app package ready to run as a container.

Log
An app's diary of what it did and what went wrong.

MQTT
A lightweight messaging system for smart devices.

Network policy
A rule saying which apps may talk to which.

Node
One computer in the cluster.

Pod
A running copy of an app (one or a few containers) on a node.

Restore test
Bringing a backup back into a test area to prove it works.

RTSP
A standard way for cameras to send live video.

Single sign-on (SSO)
Signing in once to use several apps.

Snapshot
A quick "undo point" for a disk, kept on the cluster.

Tunnel
A private line the cluster opens out to Cloudflare, so nobody has to connect in.

Two-factor sign-in (2FA)
Password plus a one-time code from your phone.

VLAN
A separate network on the same router, such as the smart-device network.

Volume
An app's own disk, managed by Longhorn.